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Professor John R. Griffiths
Editor-in-Chief, *NMR in Biomedicine*

Dear Professor Griffiths,

I am pleased to submit “Boundary-railing of conventional NLLS fits as an assumption-free pseudo-diffusion identifiability diagnostic in IVIM MRI” for consideration in *NMR in Biomedicine*.

What distinguishes this manuscript is the strength of its replication. The central finding — that a conventional box-constrained least-squares IVIM fit drives the pseudo-diffusion coefficient D^* onto a parameter bound in a large fraction of voxels — is established not once but across four cohorts and two independent implementations. On the open OSIPi in-vivo abdomen scan the railed fraction is 54.7% of high-SNR voxels (95% CI 52.2–57.1); an independent clean-room reimplementation, sharing no code with the original, reproduces a previously reported figure (54.2%, 95% CI 52.0–56.4); the effect persists over the full abdominal ROI (47.8%); and it recurs on an entirely separate organ, scanner manufacturer, and site — TCGA-LIHC liver DWI — at 43.7% (clean 4-*b* scheme) and 73.4% across three subjects (sparse 3-*b* scheme), the sparser perfusion sampling railing more, exactly as D^* identifiability predicts. Pre-registered replication thresholds, fixed before analysis, are cleared by every cohort.

This robustness is possible because the statistic assumes nothing: railing is read directly off the optimiser’s output and needs no ground truth and no noise model. That is the deliberate re-aim of the work relative to an earlier, calibration-led version. By promoting an assumption-free, cross-cohort-replicated phenomenon to the primary claim, the manuscript answers the methods objection that a calibration verdict is only as trustworthy as the simulator behind its reference truth — the railing result cannot be overextended in that way.

The calibration ruler is kept but scoped honestly: evaluated only on synthetic ground truth, reported conditionally on the true D^* regime and under a stated Cramér–Rao convention (high- D^* under-coverage of 0.63; a shape-aware quantile interval and an amortized posterior as the resolution). The earlier dramatic marginal headline is dropped and shown to be an artefact of an undocumented SD convention, now fully documented along with all dataset identifiers, fitting and training detail, and the Cramér–Rao approximation.

Every number traces to a single seeded reproduction that regenerates with one command on openly licensed data (OSIPi TF2.4, Zenodo 14605039; TCGA-LIHC via The Cancer Imaging Archive), so the replication can be re-run by any reader. The manuscript fits both principal categories of *NMR in Biomedicine* — an original in-vivo IVIM study and a methodological contribution — and is complementary to learned-uncertainty frameworks for IVIM such as Casali et al. (2026, this journal).

The manuscript is original, has not been published previously, and is not under consideration elsewhere. The author declares no competing interests, takes full responsibility for the integrity of

the work, and includes a declaration of AI-assisted writing. I would be glad to suggest reviewers and to answer any questions.

Sincerely,

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